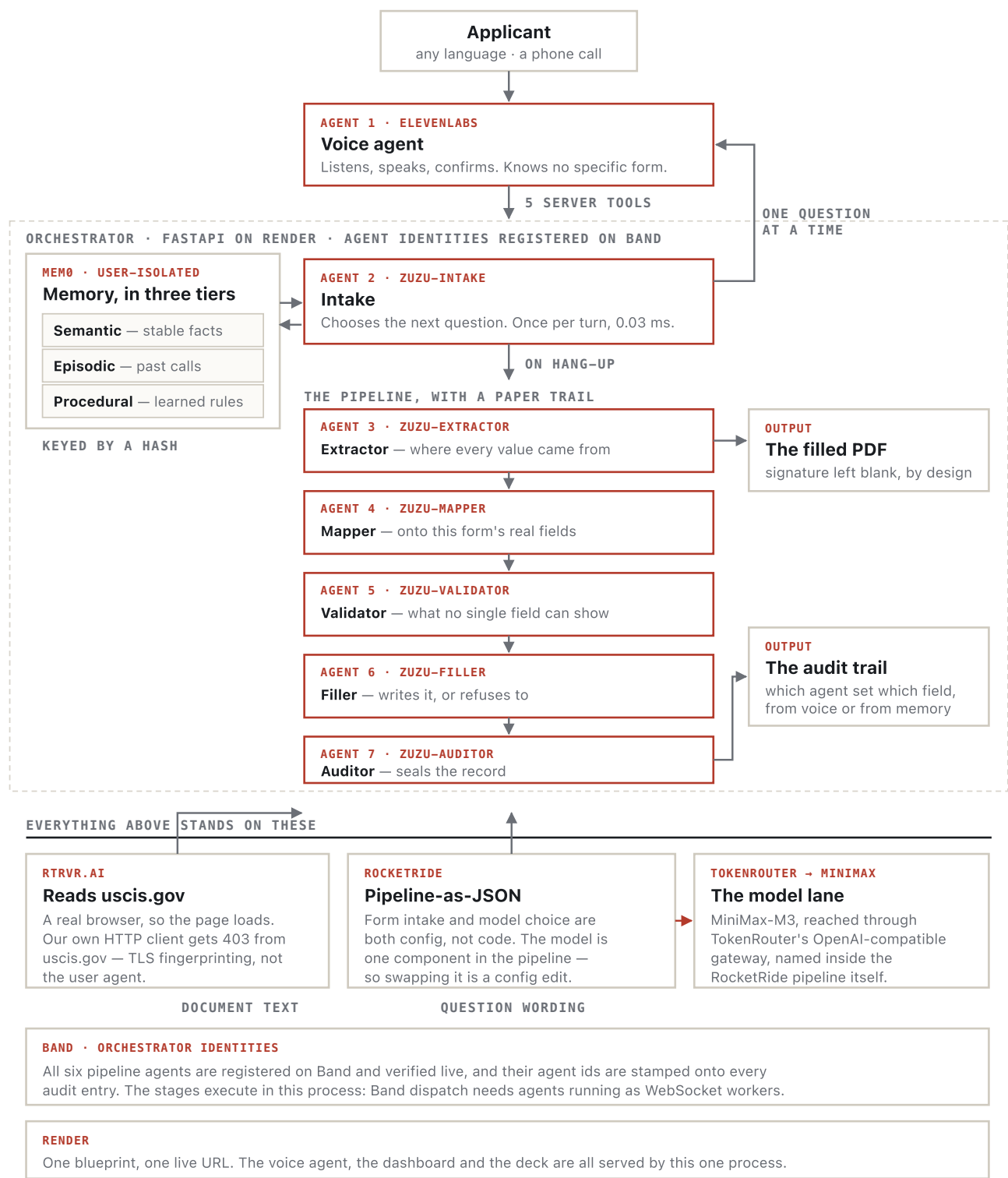


What works, and how it fits together.

Zuzu fills U.S. immigration forms by voice. You call a number, speak your own language, and answer one plain question at a time. This is what is built, what is verified, and what each piece actually does.

FORMS SUPPORTED 12	QUESTIONS MAPPED 1,900approx	AGENTS 7
TESTS 84passing	STATUS Live	

Architecture



SEVEN AGENTS. ONE VOICE AGENT IN FRONT; SIX REGISTERED BAND IDENTITIES BEHIND IT.

The strongest things about it

A form is data, not code

Twelve USCIS forms, roughly 1,900 questions, and not one of them written by hand. Zuzu fetches the fillable PDF, extracts its AcroForm inventory, and writes the questions from the document's own screen-reader labels. Adding a thirteenth form is a request, not a release. Field names are always extracted from the document and never recalled by a model — a hallucinated field name is an application that silently loses half the answers.

The voice agent knows nothing about any form

It has five server tools and no knowledge of the I-765. It asks what the orchestrator hands it. That is what lets one agent drive twelve forms, handle a mid-call switch from a work permit to citizenship, and carry the answers already given across to the new form.

Memory has three tiers, and you can see into all of them

Semantic facts, episodic call history, and procedural rules learned about the caller are stored separately, keyed by a hash of the caller id. A returning caller answers four questions instead of thirty-three. Because the tiers are separate, someone can delete their call history and keep the profile that saves them an hour.

The audit trail is the product

In a legal-filing domain the trail matters as much as the filing. Every entry records which agent set which field, from voice or from memory, and every finding. A completed demo call produces 96 entries across all six pipeline agents. When a form comes back rejected months later, somebody has to be able to work out why.

It refuses rather than guessing

The Filler will not write a form when a required answer would not survive contact with the page. Say "Texas" and the state box — which accepts only the export codes in the document — would silently discard it; Zuzu blocks and says so instead of handing back a form that reports success with an empty line on it. The same rule covers unreadable dates and model-written question wording that arrives malformed.

It was built against the real document

The official I-765 disagreed with our own written spec four times: the order of the checkboxes in Part 1, three questions that do not exist on this edition, a hybrid AcroForm/XFA structure that makes a naive fill open blank, and per-field checkbox export values with leading and trailing spaces in their names. Every one of those would have shipped as a silent wrong answer.

The seven agents, and what each one does

AGENT	RUNS	JOB
Voice agent ElevenLabs	Every turn	The whole interface. Listens in the caller's language, reads values back for confirmation, and asks exactly the question it is handed. Holds no knowledge of any specific form.
Intake Zuzu-Intake	Every turn	Decides what to ask next, honouring fields already known from memory and skipping questions gated on an answer that rules them out. Records each choice on the trail.
Extractor Zuzu-Extractor	On hang-up	Establishes provenance: for every value, whether it came from the applicant's voice, from memory, or from a document.
Mapper Zuzu-Mapper	On hang-up	Places each value on this form's real fields. A value with no destination on this form is not an error, but it is not counted as filled either.
Validator Zuzu-Validator	On hang-up	The cross-checks no single field can show: arrival dated before birth, a student category with no SEVIS number, a value that will overflow its printed box.
Filler Zuzu-Filler	On hang-up	Writes the official AcroForm, or refuses. Handles decryption, the stale XFA overlay, and per-field checkbox export values. Signature fields are left blank by design.
Auditor Zuzu-Auditor	On hang-up	Seals the record and returns it: every agent action, every finding, in order.

Memory

TIER	HOLDS	WHY IT IS SEPARATE
Semantic	Name, date of birth, country of citizenship, eligibility category	Facts that outlive the call. This is what prefills the next form.
Episodic	Which form, how many answers, whether it finished, when	Lets Zuzu say "last time we filed your renewal" rather than greeting a returning caller as a stranger. Deletable on its own.
Procedural	Speak Spanish with this applicant; no SSN, stop asking; category is (c)(3)(B), confirm rather than re-derive	How to serve this person. Learned once, applied on every later call.

All three are held in mem0 under a key derived by SHA-256 from the caller id, so the store never holds a raw phone number. Sensitive values are not persisted unless explicitly opted in. An applicant can delete one tier and keep the others.

What each integration actually does

TOOL	VERIFIED	ROLE
ElevenLabs	Live	The entire interface. Five webhook server tools; a conversation-initiation webhook supplies what memory already knows.
mem0	Live (writes)	Semantic, episodic and procedural memory, isolated per caller. Reads are currently blocked by an exhausted free-tier quota; the service says so on screen rather than pretending the caller is new.
Band	Live	Agent identities for the six pipeline roles, all confirmed registered through Band's API, with their ids stamped on every audit entry.
RocketRide	Live	Pipeline-as-JSON. Both document intake and model selection are declared as pipeline config rather than written in code, so changing the model is a config edit.
TokenRouter	Live	OpenAI-compatible gateway carrying MiniMax-M3, named inside the RocketRide pipeline definition. Used to improve question wording, never to invent a field name.
rtrvr.ai	Live	Reads uscis.gov, which returns 403 to our HTTP client regardless of headers — the block is TLS fingerprinting. rtrvr drives a real browser and returns the page, which is where the document requirements come from.
Render	Live	One blueprint, one URL. API, dashboard and deck all served by the same process.

Where it is honest about the limits

- **Band dispatch.** The six identities are registered and verified, but the stages execute in this process. Dispatching over Band's transport needs agents running as WebSocket-connected workers — a different shape of deployment, not a missing line of code. The seam is built and named.
- **Cerebras.** The account returns 402 and the model our docs named does not exist on it. It never ran, so that track is not claimed. The inference lane is provider-agnostic.
- **Email delivery.** The pipeline runs; Gmail has no credentials configured, so the code returns `delivered: false` rather than a green tick nobody can find in an inbox.
- **Zuzu fills to the review step.** It does not submit, does not pay fees, and cannot sign. The applicant reviews, signs and files.

Recently hardened

An adversarial audit — agents mapping the system and hunting defects across four lenses, with every finding handed to an independent agent told to refute it — produced 18 confirmed defects, all now fixed and covered by tests. The ones that mattered most:

- Sessions opened without a caller id all shared one memory bucket, so one applicant's details could be read back to the next. Anonymous sessions now get no memory in either direction.

- The completed PDF — which carries an SSN — was served to anyone who could guess a session id. Links are signed and expiring.
- A value the page could not accept was dropped while the form still reported complete.
- “Delete my data” could report success while deleting nothing. It refuses now.
- Applicant answers reached the dashboard unescaped; a stored payload executed.